TECHNICAL PRESENTATION

Lumo AI

A local AI assistant that works without the internet



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THE TASK

An assistant without a cloud server

Problem

Cloud assistants send your conversations to external servers, stop working without a connection and need a subscription.

Goal

Run a language model right on the laptop and give it a friendly interface with chat history.

Result

A working web app: three modes, images, saved chats, fully offline on an Apple Silicon MacBook Air.

FEATURES

What Lumo can do

Chat

Structured answers: short first, then step by step

Write

Emails, posts and essays in the requested tone and length

Translate

Russian ↔ English, only the translation

Images

A screenshot of a test or task: the model reads it and answers

Chat history

Chats are saved, the model remembers the last 20 messages

“Think” mode

The model reasons before answering: slower, more logical

Offline

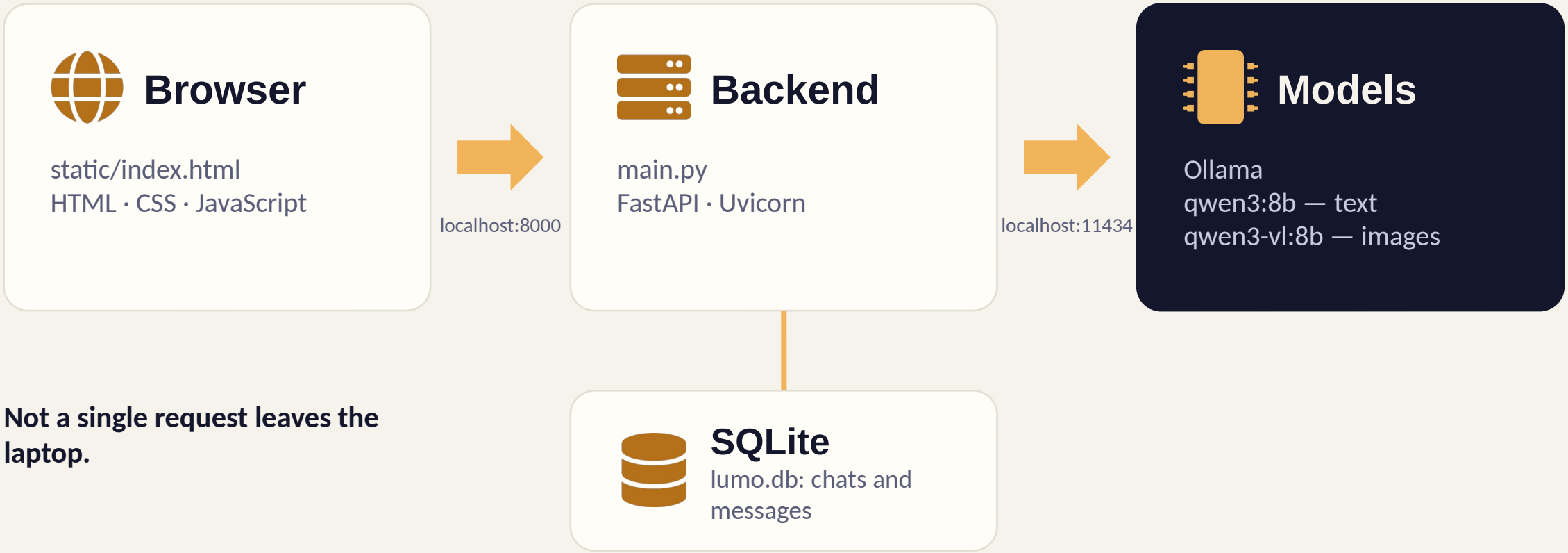
No external requests at all, even the fonts are system ones

Two languages

Russian and English interface, dark theme

ARCHITECTURE

Three layers, all on one computer



Not a single request leaves the laptop.

STACK

Technologies

Layer	Technology	Why
Language model	Qwen3 8B	Good at Russian and English, fits into 16 GB of memory (≈5 GB on disk)
Vision model	Qwen3-VL 8B	Reads text and tasks from screenshots (≈6 GB on disk)
Model runtime	Ollama	Local model server with a simple HTTP API, optimised for Apple Silicon
Backend	Python, FastAPI, Uvicorn	Async API, automatic docs at /docs, data validation with Pydantic
HTTP client	httpx	Async requests to Ollama with timeouts and error handling
Storage	SQLite	Built into Python, no separate database server needed
Interface	HTML, CSS, JavaScript	No frameworks, no build step, no external CDNs: works offline

REQUEST FLOW

The path of one message

1 Interface

sends POST /chats/{id}/ask: text, images and the think flag

2 Backend

loads the last 20 messages of the chat from SQLite

3 Backend

adds the mode prompt and picks the model: qwen3-vl if there is an image

4 Ollama

generates the answer with the chosen model

5 Backend

saves the question, images and answer, sets the chat title

6 Interface

shows the answer with Markdown formatting

MODEL SETTINGS

Each mode has its own prompt and temperature

Mode	Key points of the system prompt	Temperature
Chat	Reply in the user’s language, short answer first, then step by step; don’t invent facts	0.4
Write	Follow the format, tone and length; return only the finished text	0.7
Translate	Russian → English, everything else → Russian; translation only	0.1

Temperature: lower means more precise and predictable, higher means more varied.



The think parameter. Qwen3 can reason before answering. It is off by default for speed and is turned on with the “Think” checkbox in the interface.

DATA STORAGE

Two tables in SQLite

chats

```
id  INTEGER  PK
title  TEXT
mode  TEXT
created_at  TEXT
```

1:N

messages

```
id  INTEGER  PK
chat_id  →  chats.id
role  user | assistant
content  TEXT
images  TEXT (JSON)
created_at  TEXT
```

Images are stored as base64 inside the message. The lumo.db file lives in the project folder and is not committed to Git. The chat title is the first 40 characters of the first message.

REST API

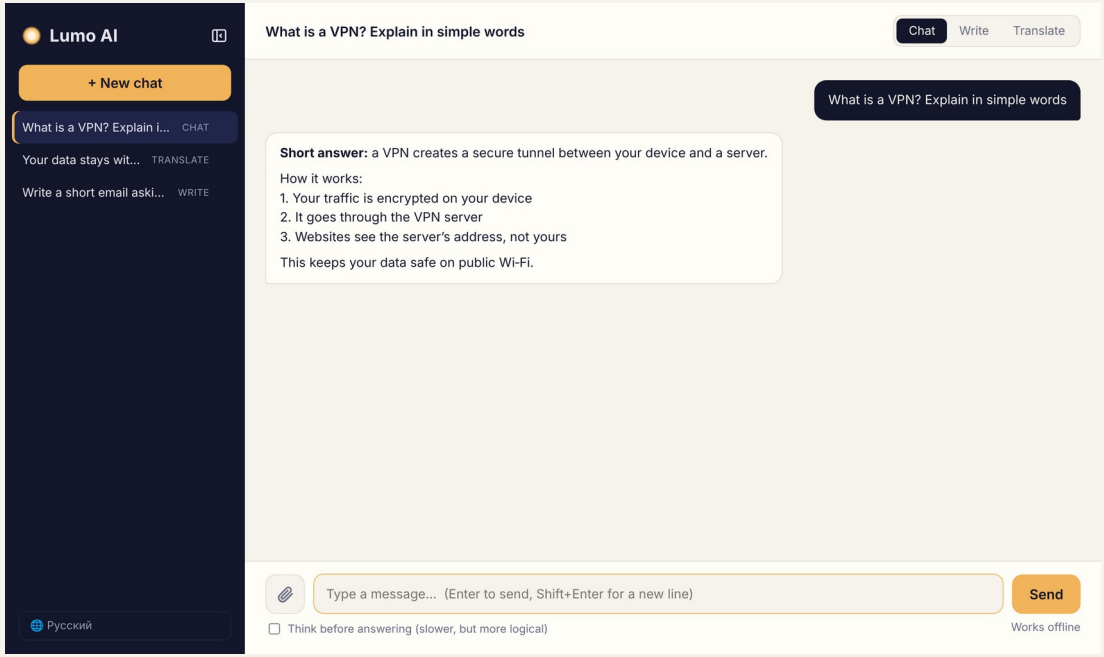
Backend endpoints

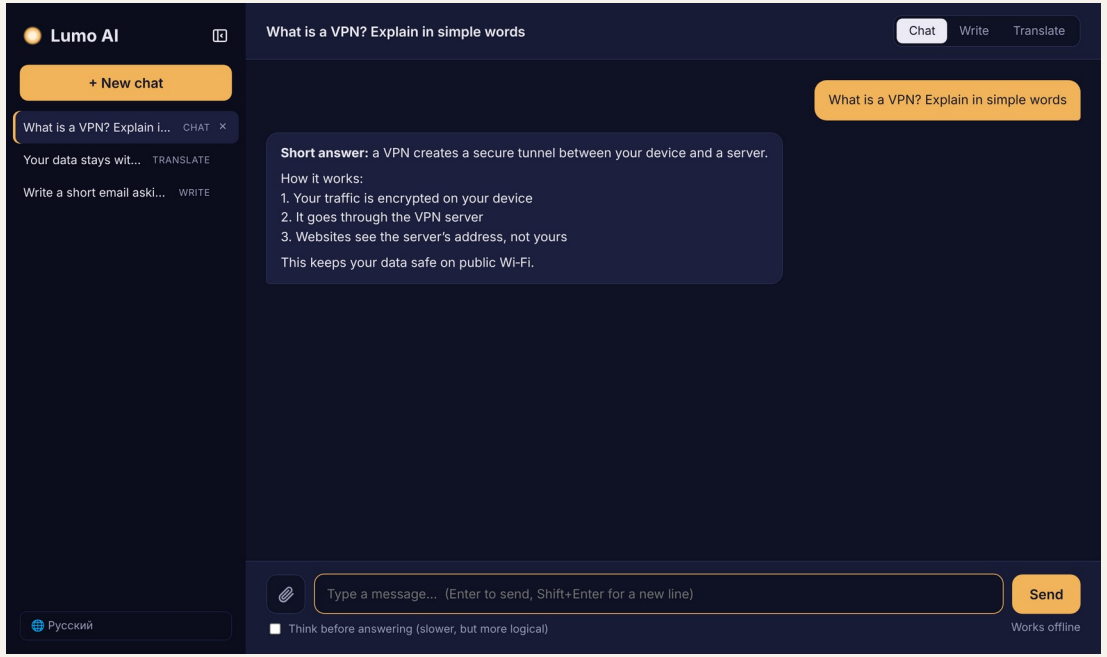
Method	Path	Purpose
POST	/ask	One-off question, not saved
POST	/chats	Create a chat with the chosen mode
GET	/chats	List chats
GET	/chats/{id}	A chat with its full history
POST	/chats/{id}/ask	Message in a chat: text, images, think
PATCH	/chats/{id}	Change the chat's mode
DELETE	/chats/{id}	Delete a chat

Interactive documentation is generated automatically: http://127.0.0.1:8000/docs

INTERFACE

Light and dark theme





- Chat list with a collapsible sidebar
- Chat / Write / Translate modes and image attachments
- Built-in presentation viewer
- Russian and English, dark theme, mobile layout

REQUIREMENTS

What it runs on

16 GB

of RAM: the test MacBook Air

≈11 GB

on disk: qwen3:8b (≈5) and qwen3-vl:8b (≈6)

0

external requests while working

macOS on Apple Silicon, Python 3.9+, Ollama. The internet is only needed to install the models and libraries.

ROADMAP

Plans

Streaming answers

Text appears word by word

Next step



Mac app

Packaged with Tauri



Your documents

Local RAG over PDF and DOCX



Password

Access from other devices



Thank you for your attention!

Any questions?